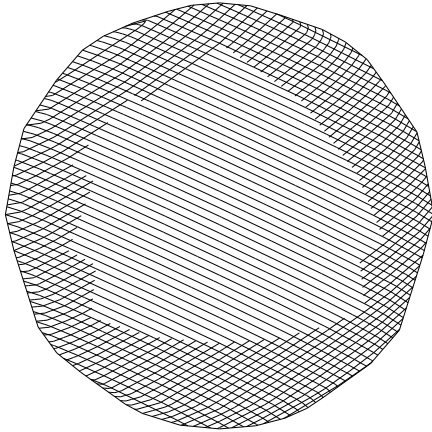


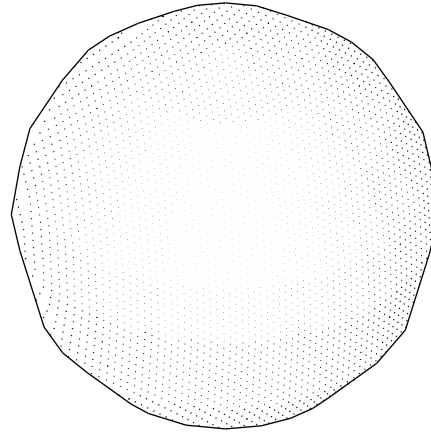
The receiver gives the figure its face

One geometric carrier; line engraving, discrete stippling and coherent phase response are different readings.

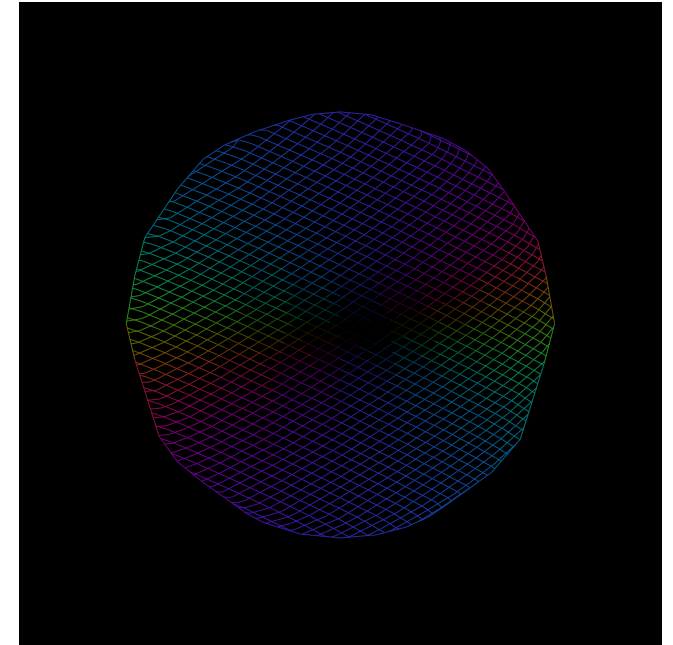
Field contours and foreshortening



Intersections of the same level families



The complex current, read on black



$(K, Z, U) \rightarrow R \rightarrow (\text{visible fibre}, \chi_0, \chi_1, P) \rightarrow \text{engraved face}$

The geometry is upstream of the texture.

Exact rational stations and triangular incidence define the finite carrier. Hatches are level traces of supplied receiver fields. The dot pattern uses intersections of those traces; no random seed places structure. The visible mesh is a finite realization of the rational sphere chart.

Method recovered. Fiziko varies stroke width, generator families and visibility to engrave curved objects; vintage-latex separates contours, texture and labels. These are original Typst packages using the existing Holonic receiver laws. White paper is fixed; phase frames are pure black. Sources: Fiziko, vintage-latex.

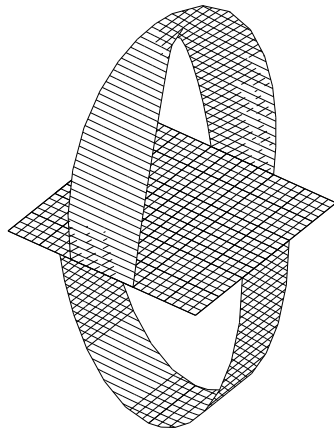
The image keeps a source behind it.

The packet retains complex vertices, currents, face addresses, the receiver and display error. Opaque-surface visibility is decided before paint. A black or unmarked region can be a zero response, an occlusion or a receiver omission; these remain different source records.

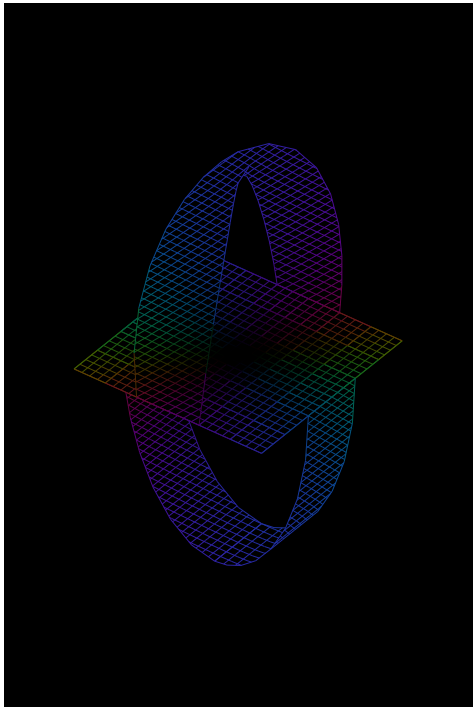
Curvature, folding and orientation have different sources

Curved rational ribbon charts preserve their attachments; the receiver carries their local current into an image.

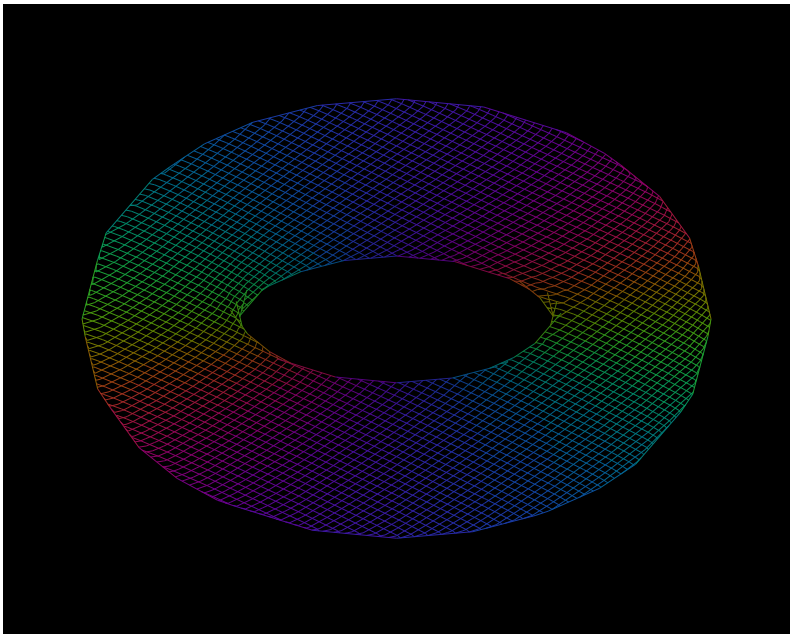
Möbius shorts · monochrome



The same packet geometry · phase



A closed orientable carrier



$$\chi(\text{torus}) = 0, \quad \chi(\text{shorts}) = -1$$

The torus is orientable. Shorts have one boundary and an orientation-reversing loop. Their appearance under projection proves neither fact; the incidence witness does.

$$\varepsilon_g = -h_{fe}h_{ge}\varepsilon_f, \quad \prod_{\gamma}(-h_{fe}h_{ge}) = -1$$

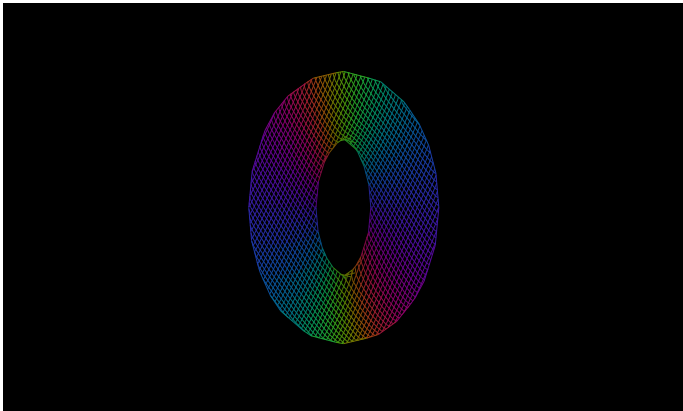
Exact finite construction. The ribbon uses a rational transverse frame whose squared length is $1 - \frac{3\sin^4 t}{4}$. Its width therefore varies along the curved chart. The earlier unit-normal ribbon and its chosen $\frac{1}{12}$ shell offset remain in the companion atlas.

Receiver distinction. A polyhedral source has curvature concentrated at its edges. A projection may flatten a curved source. Intrinsic nonorientability is a separate failure of global orientation, while local transport and receiver-relative orientation remain available.

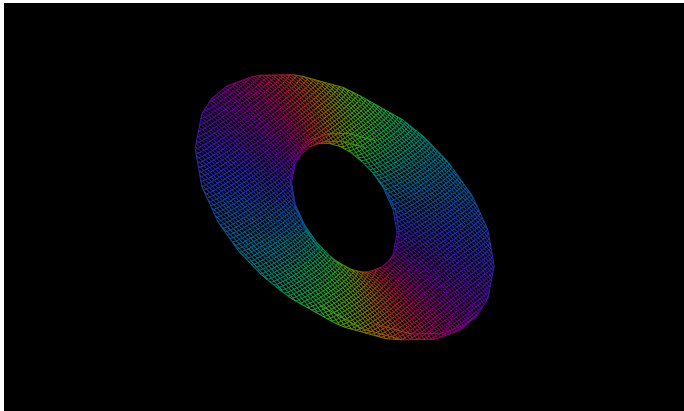
A complete complex current drives the motion

An exact holomorphic Euler–NS source on $\mathbb{C}^3$; no Fourier-reality condition discards the imaginary field.

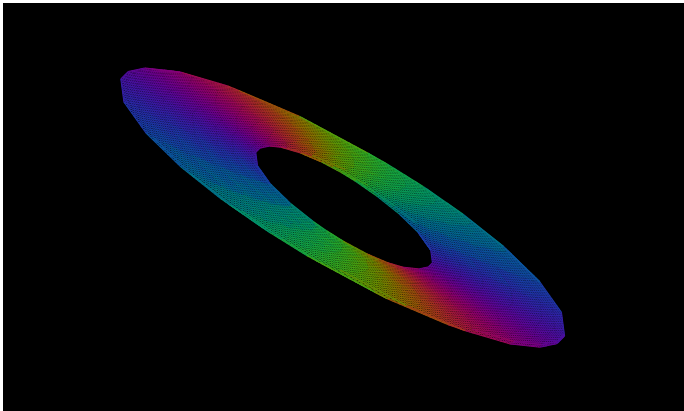
$\tau = 0$



$\tau = \frac{1}{2}$



$\tau = 1$



$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & i \\ 1 & i & 0 \end{pmatrix}, \quad A^3 = 0, \quad U(Z) = AZ, \quad P(Z) = -\frac{1}{2}(Z_1 + iZ_2)^2$$

$$F_\tau = I + \tau A + \frac{\tau^2}{2} A^2, \quad F_{-\tau} F_\tau = I, \quad \det_{\mathbb{C}} F_\tau = 1$$

$$(U \cdot \nabla)U = A^2 Z = -\nabla P, \quad \Delta U = 0, \quad \nabla \cdot U = 0$$

The two fields interact.

$$a_t + (a \cdot \nabla)a - (b \cdot \nabla)b = \nu \Delta a - \nabla p$$

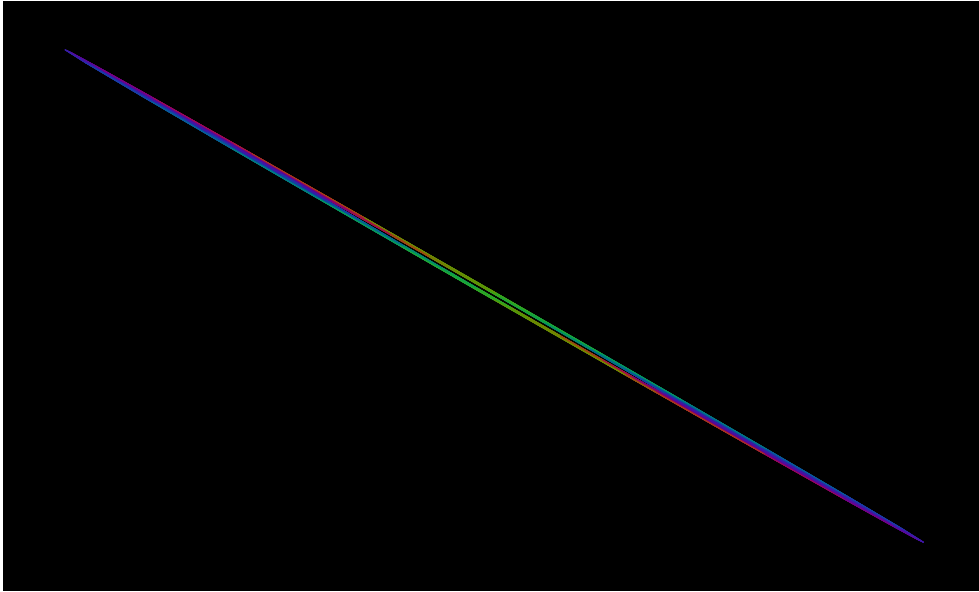
$$b_t + (a \cdot \nabla)b + (b \cdot \nabla)a = \nu \Delta b - \nabla q$$

Proved-derived. $U = a + ib$, $P = p + iq$; advection is complex-bilinear. The holomorphic differential is $\Delta = \sum_j \partial_{Z_j}^2$. This polynomial field satisfies the displayed equation for every declared viscosity. The receiving image is a real projection of the full complex flow; the camera and drawing scale are held fixed.

A singular image can have a regular source

Keep the complex fibre when a receiver loses directions. A second receiver can expose them again.

Real receiver · $\tau = \frac{7}{5}$

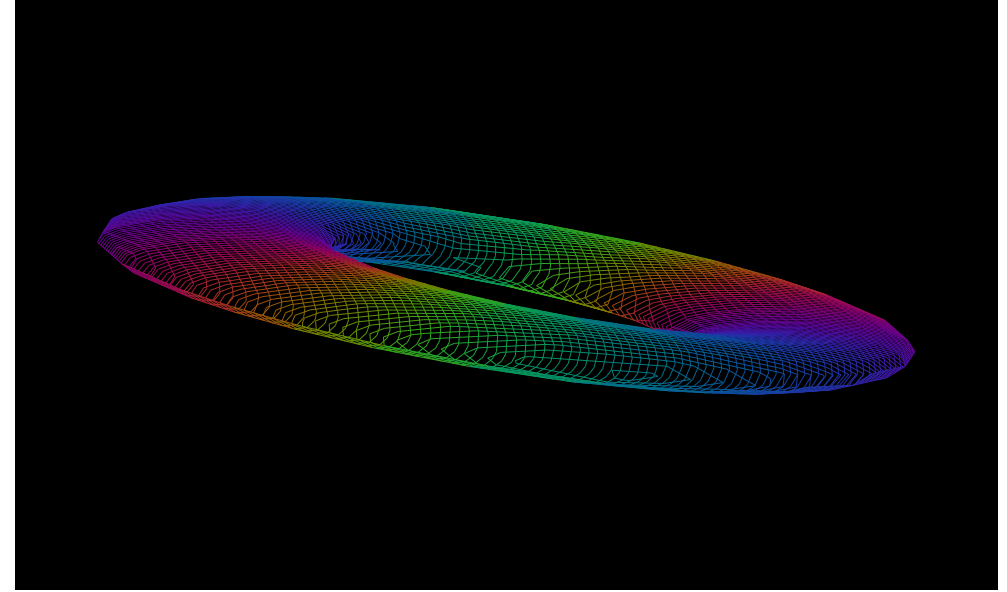


$$R_0 F_\tau = \operatorname{Re} F_\tau, \quad \det_{\mathbb{R}}(R_0 F_\tau) = \left(1 - \frac{\tau^2}{2}\right)^2$$

$$R_{a,b}(Z) = a \operatorname{Re} Z + b \operatorname{Im} Z, \quad (a, b) = \left(\frac{3}{5}, \frac{4}{5}\right)$$

Proved-derived. At $\tau^2 = 2$ this real receiver has rank one, while F_τ is invertible. The picture uses rational $\tau = \frac{7}{5}$ near that exact singular cut. The source is not replaced by the nearly collapsed image; its complex coordinates and current are retained.

Changed receiver · same source and clock

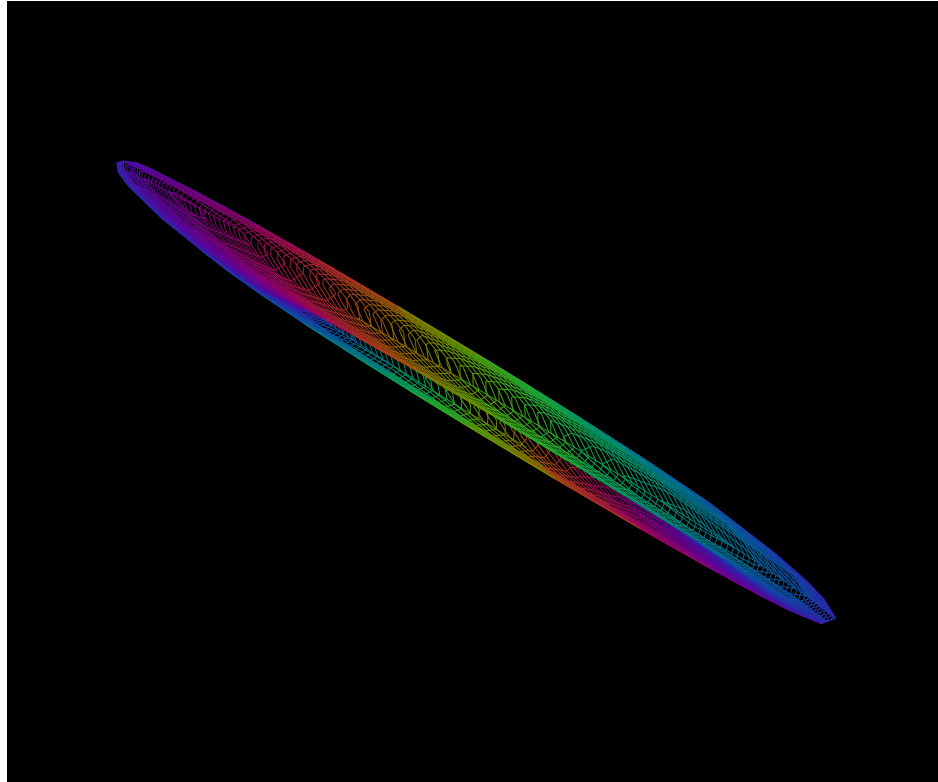


Actual folded source. On the real slice, $a = (x_3, 0, x_1)$ and $b = (0, x_3, x_2)$. Dropping b with the same declared real pressure drops $(b \cdot \nabla)b = (0, x_2, x_3)$. This is why the complex evolution cannot generally be continued from its present real face alone.

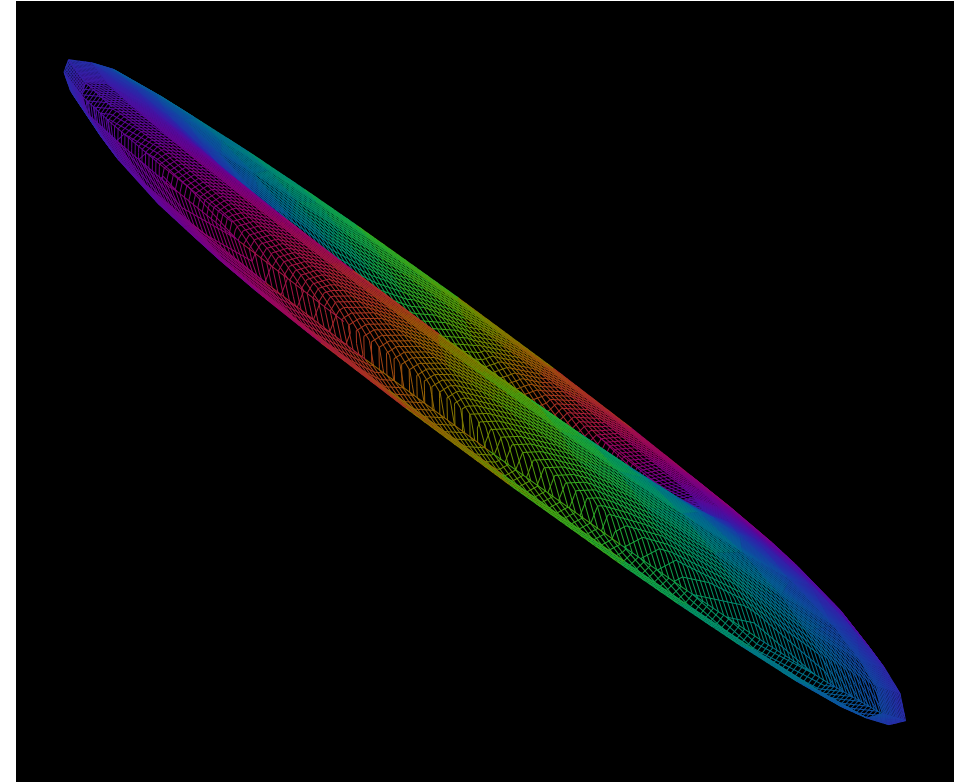
Current and curl have separate receivers

Add a declared rotational current to the same complex source family; read circulation through actual face boundaries.

$$\lambda = \frac{1}{2}, \tau = \frac{7}{5}$$



$$\lambda = 1, \tau = \frac{7}{5}$$



$$v = (1, i, 0), \quad KZ = v \times Z, \quad A_\lambda = S + \lambda K, \quad U_\lambda = A_\lambda Z$$

$$A_\lambda^2 = (1 + \lambda^2)S^2, \quad A_\lambda^3 = 0, \quad \text{curl } U_\lambda = 2\lambda v$$

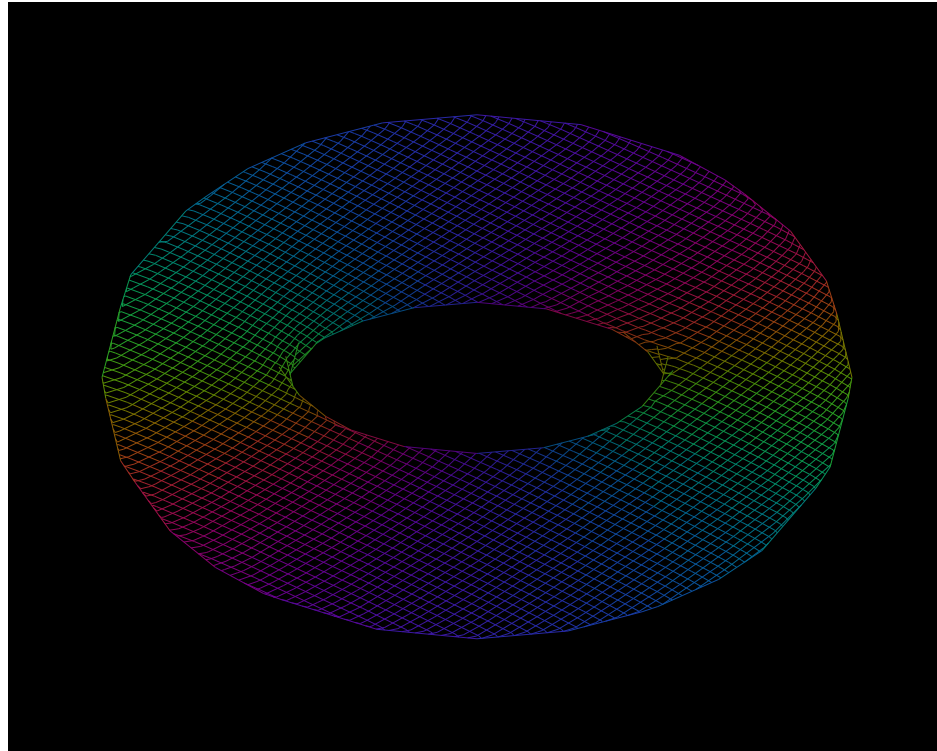
$$j_{pq} = \frac{U_{\lambda(p)} + U_{\lambda(q)}}{2} \cdot (q - p), \quad \sum_{e \in \partial f} j_e = \text{curl } U_\lambda \cdot \text{area}_f$$

Exact complex Stokes instance. S is the matrix on plate 3; $P_\lambda = (1 + \lambda^2)P_0$. On the oriented triangle $(1, 0, 0) \rightarrow (0, 1, 0) \rightarrow (0, 0, 1)$, the boundary return is $\lambda(1 + i)$. Both channels are retained. The hatches/colors read the supplied current, while this independent edge integral reads its curl. The source geometry, rotational contribution and receiver are explicit; a triangle outline alone is no flux measurement.

The cross-hatching carries a receiver differential

The line families can read current phase or distributed cross-entropy. Their source functions determine the marks.

Complex-current level families



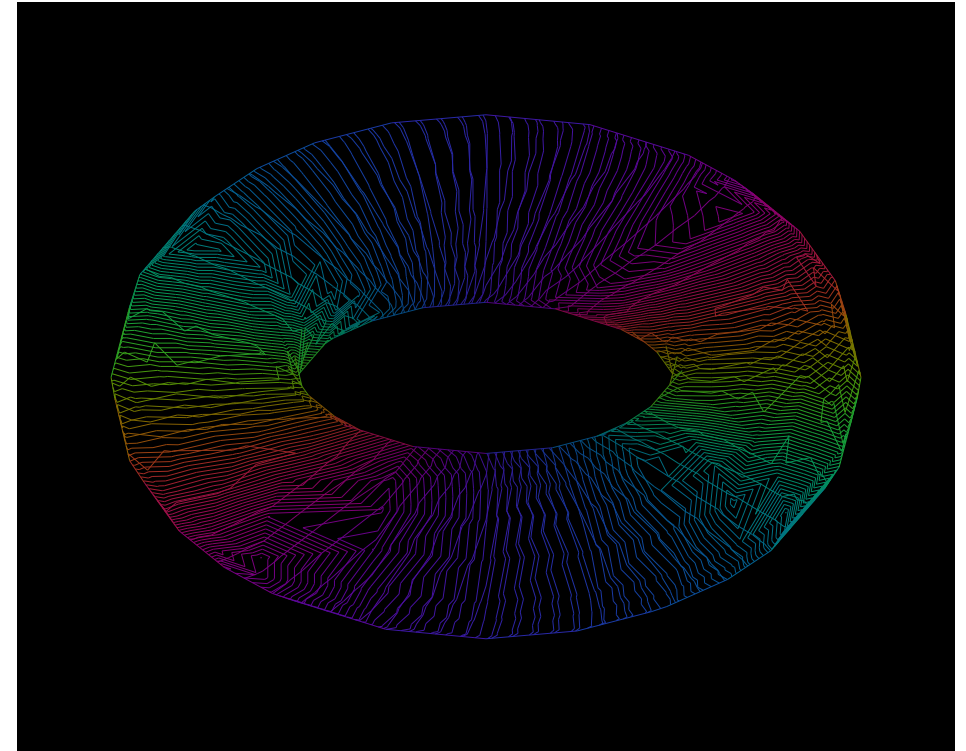
$$P = (a^2, b^2, (a+b)^2), \quad C = \frac{P}{\kappa + \sum_j P_j}, \quad \sum_j C_j + \frac{\kappa}{\kappa + \sum_j P_j} = 1$$

$$q_i = \frac{\frac{\kappa}{3} + P_i}{\kappa + \sum_j P_j}, \quad p_i = \frac{1}{3}, \quad h_i = -p_i \log q_i, \quad H = \sum_i h_i$$

$$dH = \sum_i (q_i - p_i) ds_i \quad (q = \text{softmax}(s))$$

Defined receiver. The right field uses h_0, h_1 as two level families, with the third contribution retained by the source law. Positive support makes the logs defined. Station logs carry rational enclosures; the rendered field is their declared affine triangle reading.

Distributed cross-entropy families

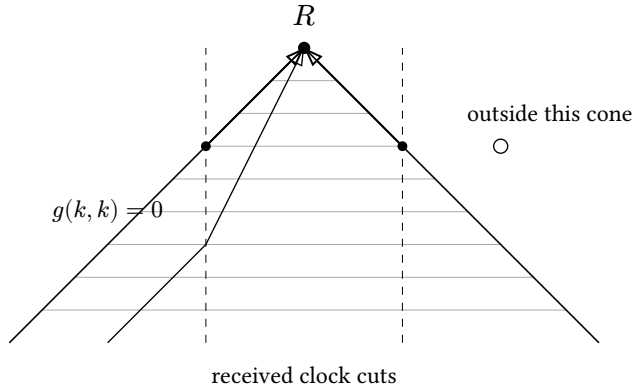


$$\int_{\gamma} d\chi = \chi(b) - \chi(a) = \Delta N + r_b - r_a, \quad |r_b - r_a| < \Delta$$

Proved-derived. Signed crossings of levels $k\Delta$ count the quantized potential flux. Endpoint remainders retain what finite line density omits. No hatch direction is assigned by semantic label, and a geometric slope is not silently renamed cross-entropy.

The receiver is situated at an event

World-lines, cone membership, orientation and ordered frame changes belong to the reading.



$$g(u, u) = -1, \quad g(k, k) = 0, \quad \omega_R = -g(k, u)$$

Declared Lorentz receiver. A time-oriented metric, observer world-line and tetrad determine the received frequency and direction. Cone membership permits causal influence; actual incidence determines whether that influence occurs.

A curved metric changes the received frequency

$$f(r) = 1 - \frac{2}{r}, \quad g = \text{diag}(-f, f^{-1}, r^2, r^2), \quad k = (f^{-1}, 1, 0, 0)$$

$$r_{\text{near}} = \frac{25}{8}, \quad r_{\text{far}} = \frac{50}{9}, \quad \omega_{\text{near}} = \frac{5}{3}, \quad \omega_{\text{far}} = \frac{5}{4}, \quad \frac{\omega_{\text{far}}}{\omega_{\text{near}}} = \frac{3}{4}$$

Exact Schwarzschild receiver instance. $G = c = M = 1$ at the equatorial slice; static tetrads are orthonormal, and the radial null ray has unit conserved Killing energy. Its two nontrivial geodesic equations cancel exactly. The frequency ratio follows from the metric and observer; no color is assigned to manufacture a redshift.

$$\nabla_a a - \nabla_b b, \quad \nabla_a b + \nabla_b a$$

Complex/GR interface. Complexifying a declared connection gives these two coupled advection terms. A holomorphic C^3 differential is a different chart from a Lorentzian space-time derivative. Their comparison retains the real restriction, tetrad, pressure/source return and any connection defect. No image supplies a stress-energy tensor or Einstein solution by appearance.

The gyroparallelogram retains the order

$$L_x = \begin{pmatrix} \frac{5}{4} & -\frac{3}{4} & 0 \\ -\frac{3}{4} & \frac{5}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$L_y = \begin{pmatrix} \frac{5}{3} & 0 & -\frac{4}{3} \\ 0 & 1 & 0 \\ -\frac{4}{3} & 0 & \frac{5}{3} \end{pmatrix}$$

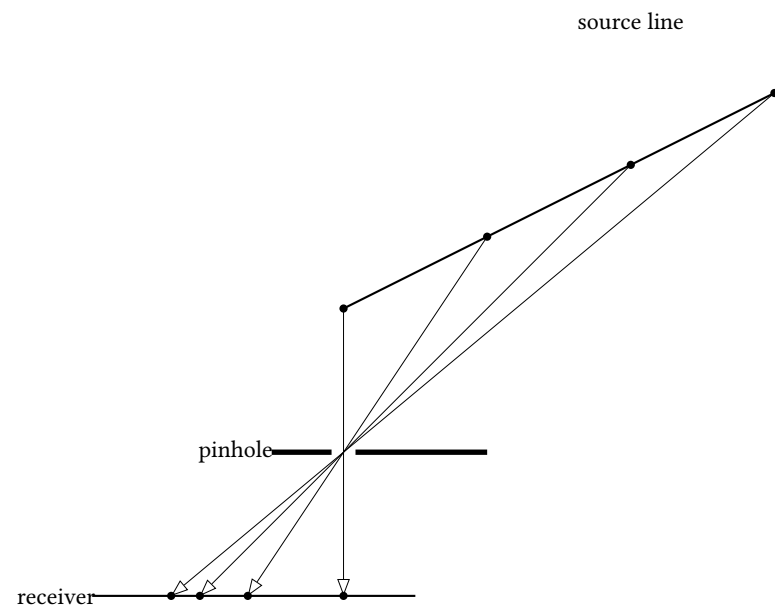
$$L_y L_x \neq L_x L_y, \quad L^\top \eta L = \eta$$

$$B_{L_y L_x e_0} L_y L_x = \text{diag} \left(1, \begin{pmatrix} \frac{35}{37} & -\frac{12}{37} \\ \frac{12}{37} & \frac{35}{37} \end{pmatrix} \right)$$

Exact rational return. The canonical return boost removes the final velocity; a spatial rotation remains. The 37 is produced by the two supplied boosts. This is a flat-spacetime frame instance; a curved GR receiver uses its supplied metric and connection.

A shadow retains a path behind its face

Projection, analytic continuation and a physical Feynman diagram have their own source maps.



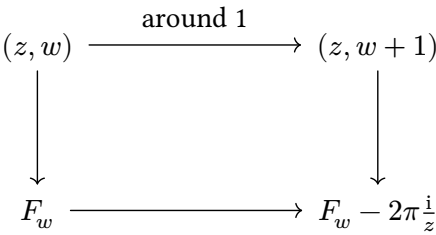
$$f(t) = \frac{t}{1 + \frac{t}{2}}, \quad \text{cr}(0, 1, 2, 3) = \text{cr}\left(0, \frac{2}{3}, 1, \frac{6}{5}\right) = \frac{4}{3}$$

Exact projective instance. A noncollapsed line retains its cross-ratio. The pinhole denominator has a pole at $t = -2$; the source point remains defined. Projection does not preserve arbitrary lengths or reconstruct its fibre.

Recovered context. The August 20 “Shadows of Holonic Interactions” deposit refers to Tufte’s physical shadows of Feynman-diagram sculptures. Our formal owner proves polarity and conjugate-line identities. Hypergeometric monodromy supplies the separate analytic example above. Actual Feynman amplitudes require their propagators, couplings and integration/regularization data; a projected drawing does not supply them.

A hypergeometric face with a retained branch

$$F(z) = {}_2F_1(1, 1; 2; z) = -\frac{\log(1 - z)}{z}$$

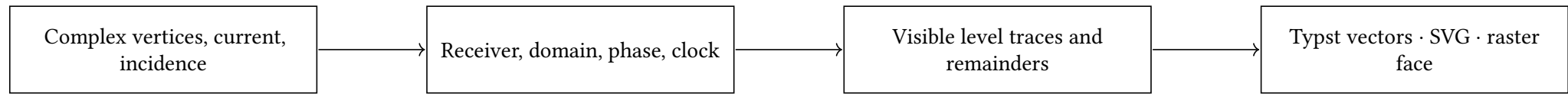


$$F_w = F_0 - \frac{2\pi iw}{z}, \quad w \in \mathbb{Z}$$

Derived continuation. A positive loop around 1 changes the logarithm by $2\pi i$. The winding integer and source branch generate the family; the endpoint z alone does not identify it. Source: DLMF 15.4.1.

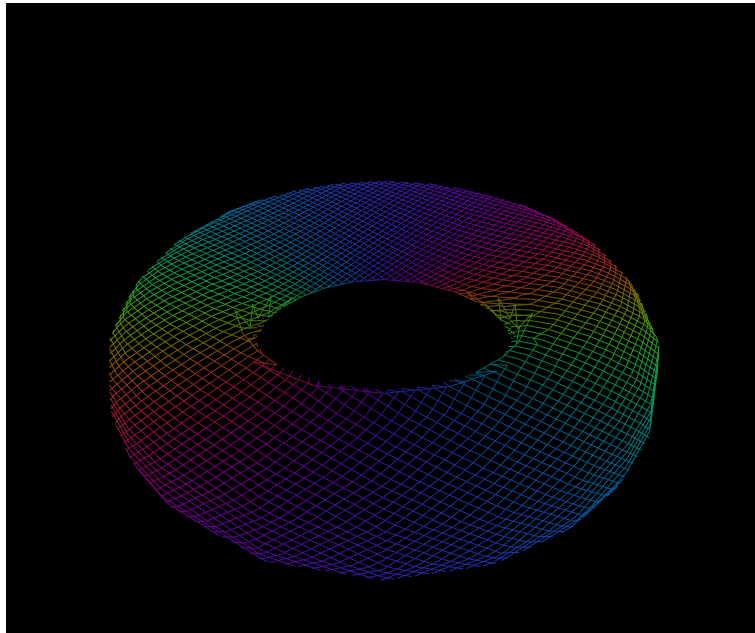
The packet carries the construction

A reusable receiver package and an engraving package replace picture-specific paint rules.



holonic-receiver

Exact rational companion: source projection, affine or reciprocal-depth visibility, level intersection, coherent primary response and enclosed entropy stations. Pure Typst helpers expose receiver maps and the softmax cross-entropy differential. The packet keeps source and omitted/degenerate populations.



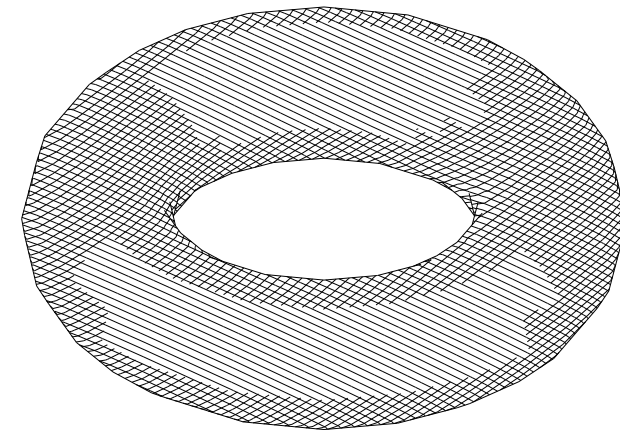
$$qT_w = U_w q, \quad rT_w = \tilde{r}U_w q$$

$$C(A) = C(-A), \quad C(A+1) \neq C(-A+1) \quad (A = 1 + 2i)$$

Future-receiver condition. A cached image is sufficient only for the receivers and continuations it preserves. A generator, current and retained fibre can describe indefinitely many later configurations without storing all pictures. ETP's substitution/congruence discipline applies to admitted passages; identical current pixels do not establish it. The companion twenty-plate composition atlas retains the prior HNN/Athena architecture and its exact examples.

holonic-engraving

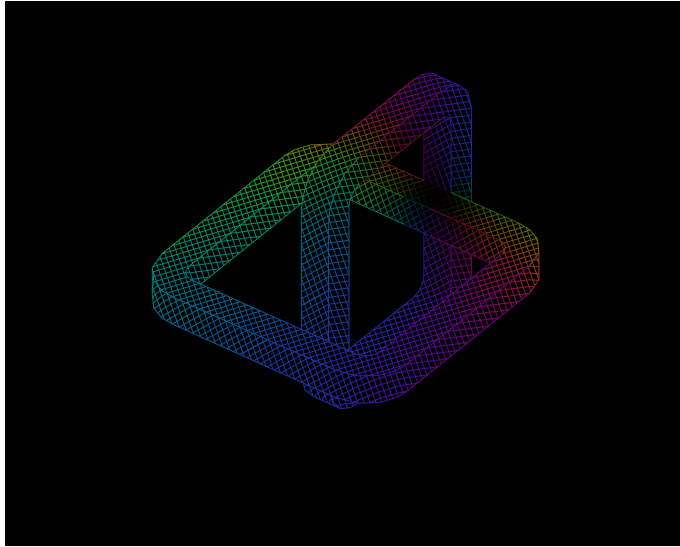
CeTZ draws supplied marks. Line width, stippling and paint are explicit display choices; clipping and source identity are already in the packet. Phase colors are transferred without a contrast boost. The monochrome mode changes paint, not the supplied geometry.



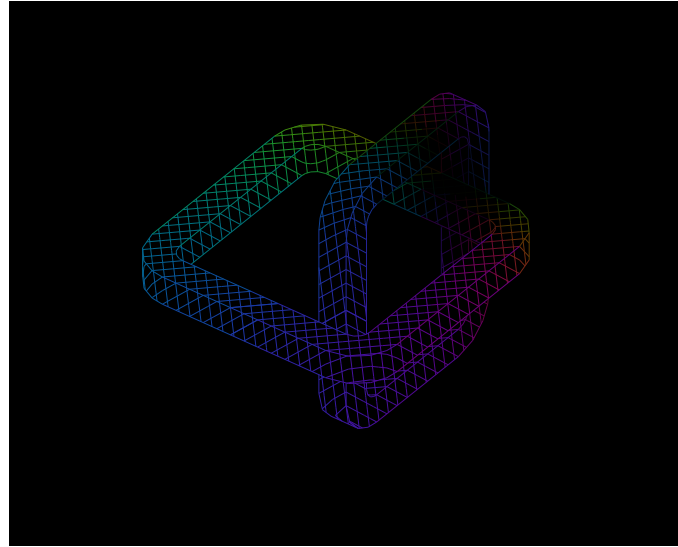
One link changes the motion of the other

Consequential boundary coupling: normal grip, tangential slip, and energy carried into the shared interface.

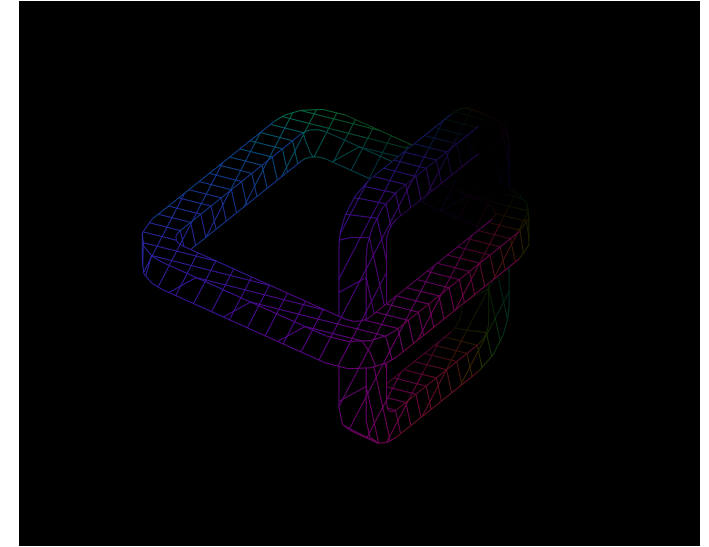
Initial slip · $a = 1$



Coupled motion · $a = \frac{1}{2}$



Reduced slip · $a = \frac{1}{8}$



$$a = e^{-2t}, \quad s = e_y + e_z, \quad v_A = as, \quad v_B = -as, \quad c_A = \frac{1-a}{2}s, \quad c_B = -c_A$$

$$f_A = -2as = -f_B, \quad E_{\text{kin}} = 2a^2, \quad P_{\text{friction}} = 8a^2, \quad Q = 2(1 - a^2)$$

A real shared patch constrains the movement.

$$g \geq 0, \quad N \geq 0, \quad gN = 0, \quad g = 0, \quad N = 1$$

Exact guided-body model. Two unit-mass rounded links have disjoint interiors and a shared square of area $\frac{1}{4}$. Opposed normal preloads are balanced by contact reactions. Guides hold their rotations; tangential motion follows unit viscous contact. This is a declared contact law, not an unconstrained elastic-knot solver.

Recovered meaning. The Turn tablet defines friction as the consequential coupling itself, with grip and slip as two faces. The positive heat return here belongs to this particular viscous constitutive law. At the displayed phases, $t = -\frac{\log(a)}{2}$; positions, forces and energy are exact ratios, rather than an interpolated animation.

Grip and turn are measured together.

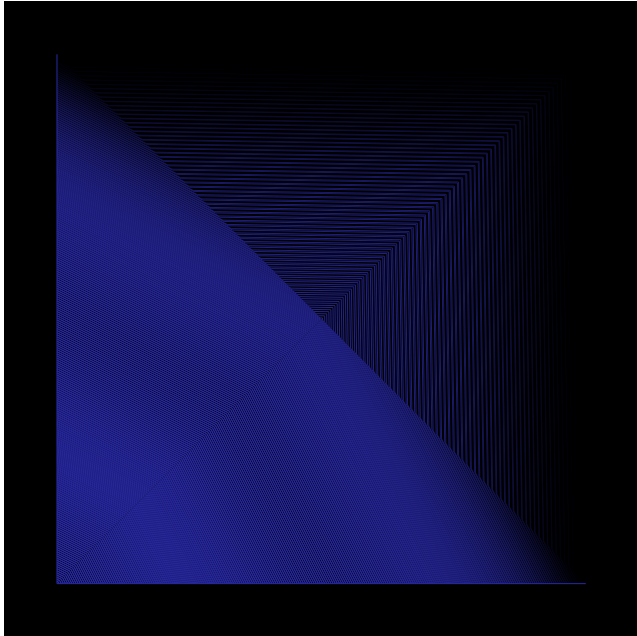
$$A_R = \frac{r \cdot F}{E_0} + i \frac{(r \times F) \cdot n_R}{E_0}$$

The body hatches read this work/turn pair with $E_0 = 2$, $F_A = -Nn + f_A$ and $F_B = -F_A$. Color follows the same primary law. The links retain linking number -1 ; no surface passes through another. The buried contact is read separately on the next plate by an interface receiver.

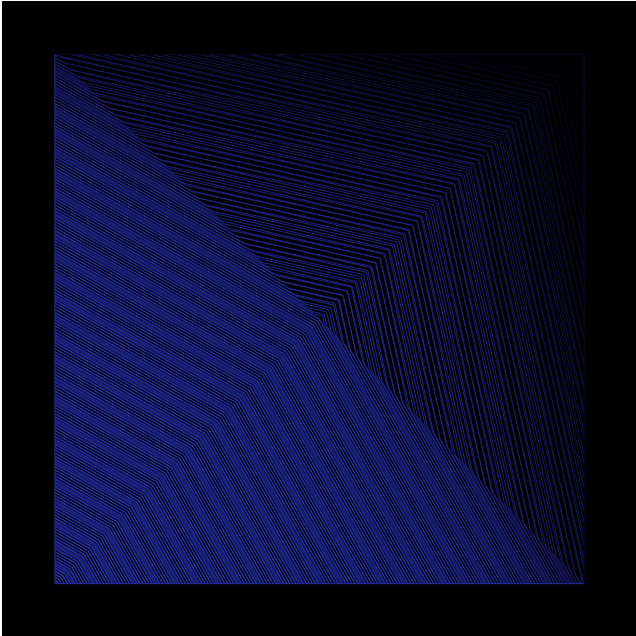
Friction drives a complex interface current

Momentum transfer and heat propagate through the actual boundary adjacency of the contact patch.

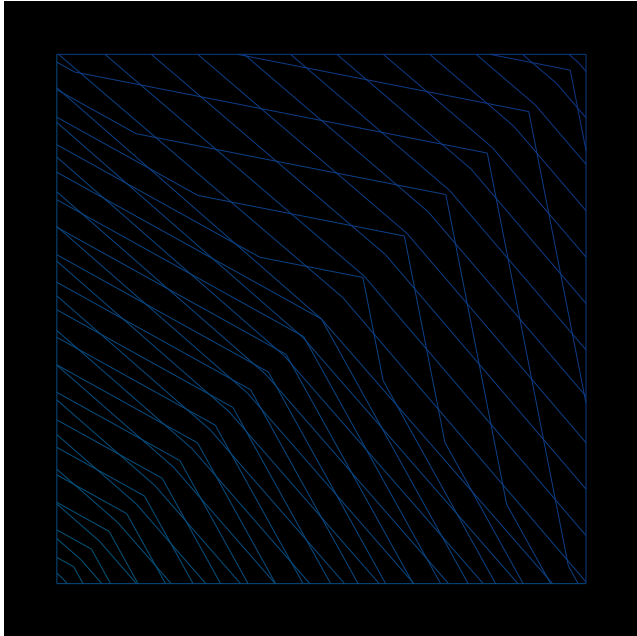
Early contact · $a = \frac{255}{256}$



Propagation · $a = \frac{15}{16}$



Distributed return · $a = \frac{1}{2}$



$$L = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}, \quad \ell = 2r = \frac{1}{2}, \quad \frac{D}{\ell^2} = 4$$

$$\dot{\theta} = -4L\theta + 8a^2e_0, \quad \dot{\Psi} = -4L\Psi + 2(1+i)ae_0, \quad \Psi \in \mathbb{C}^4$$

$$\sum_i \theta_i = 2(1-a^2), \quad \sum_i \Psi_i = (1+i)(1-a), \quad E_{\text{kin}} + \sum_i \theta_i = 2$$

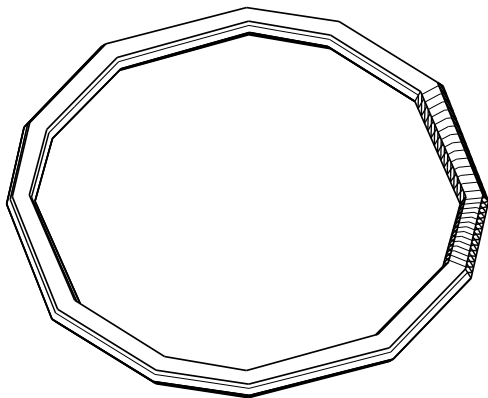
Exact finite propagation. Four boundary nodes have unit thermal capacity and unit diffusivity at their actual spacing. Frictional power enters one declared port; the operator transports it along contact edges. Ψ is the complex diffusing response to transmitted tangential force, with its sum equal to the delivered impulse. It is a history receiver, not an additional uncounted mechanical energy store.

Declared interface image. Four fan triangles interpolate the boundary readings; the center is their mean, not a fifth dynamical state. Ink reads normalized impulse rate together with stored heat. The fixed receiver aperture is the per-node quarter-area $\frac{1}{16}$, and the field-level spacing is $\frac{1}{256}$. This internal receiver has access to the contact patch; the outer views on plate 10 do not pretend to see through opaque bodies. The finite interface law accompanies the C^3 bulk constructions above without claiming a complete continuum fluid/contact solver.

The knot is retained while its measured face changes

Prime knots and an unknot are explicit closed polygonal carriers, with triangular transverse faces.

Unknot · 0₁



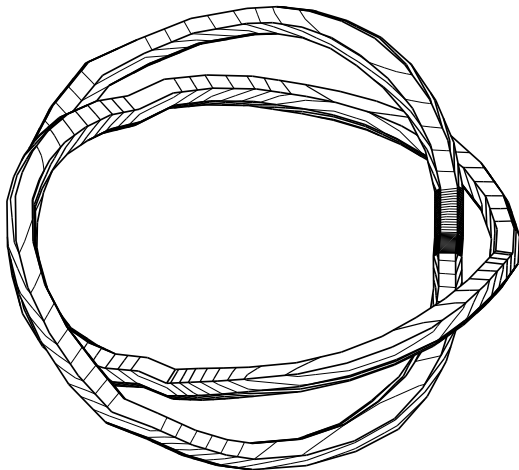
$$\Delta(t) = 1$$

convex spanning disk, $\text{closure}(\sigma_1^3)$, $\text{closure}\left((\sigma_1\sigma_2^{-1})^2\right)$

A crossing is a measured difference in depth. Established-bounded. Rational annular braid cells return exactly 0, 3 and 4 projected crossings. The generator checks every nonincident segment pair, the angular cell and the actual over/under sign. The prime labels and Alexander polynomials are standard identifications of these certified diagrams; they are not inferred from hue.

The current Lean torus-slope owner proves coprime embedded circles and the quandle owner supplies local transport laws. Ambient knot classification is carried here by the explicit standard braid diagram and exterior exact checks. Source: Knot Atlas.

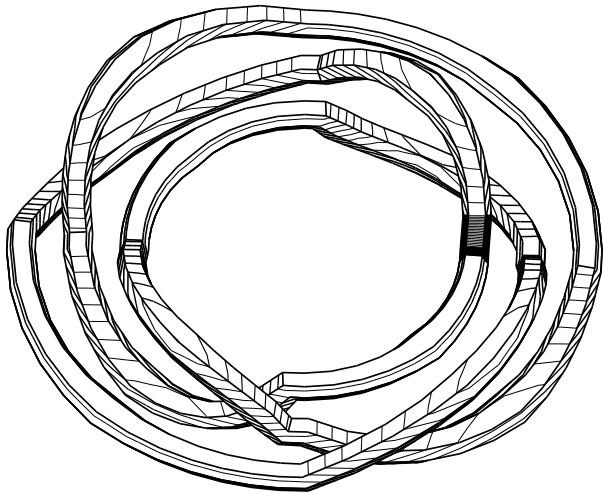
Trefoil · 3₁



$$\Delta(t) = t^{-1} - 1 + t$$

Thickness retains the actual joints. Established-bounded. Rational transverse hexagons form a closed oriented triangular tube. All triangle pairs are checked for intersections beyond their shared combinatorial face, including coplanar joints. Each component has $\chi = 0$. The affine fluid map moves entire faces exactly; it cannot undo the knot.

Figure-eight · 4₁

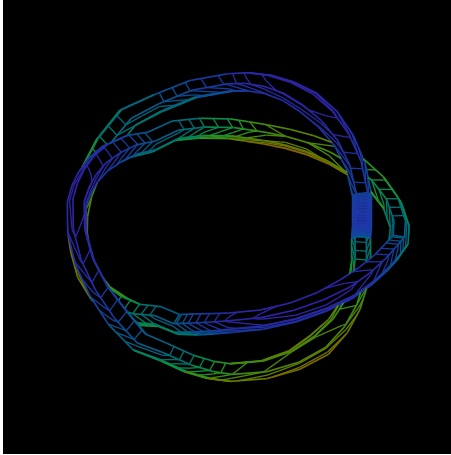


$$\Delta(t) = -t^{-1} + 3 - t$$

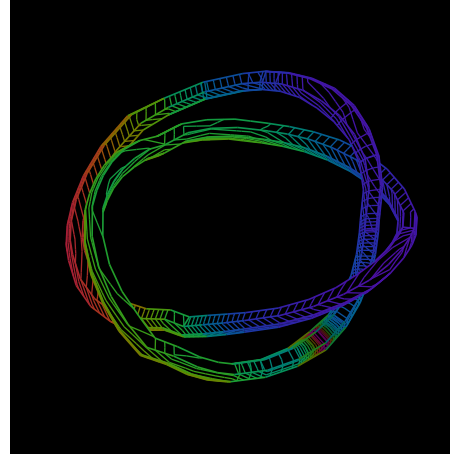
An impulse travels while the trefoil is perturbed

One causal source, two counterpropagating complex ports, a moving body, and an explicitly retained heat return.

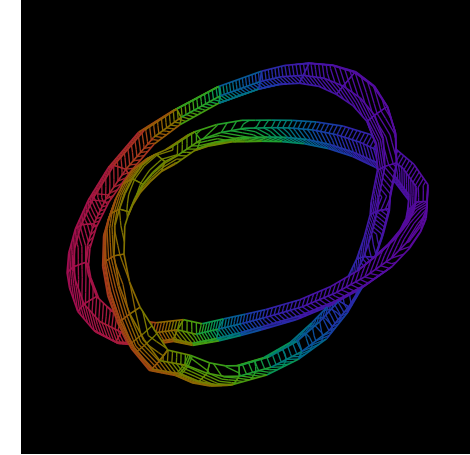
$k = 0, t = 0$



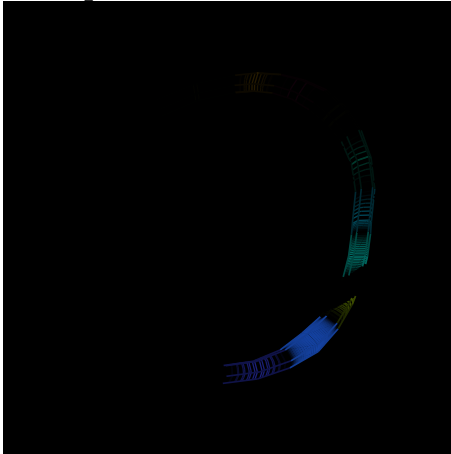
Joint face $\cdot k = 12$



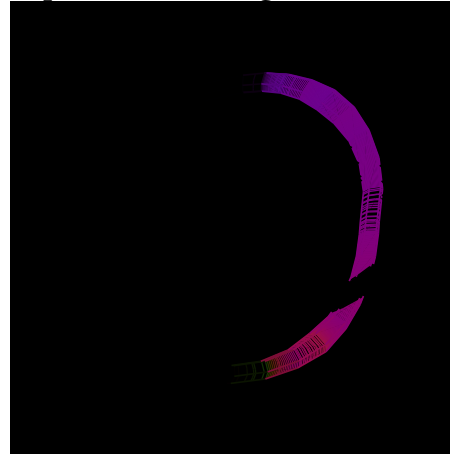
Joint face $\cdot k = 24$



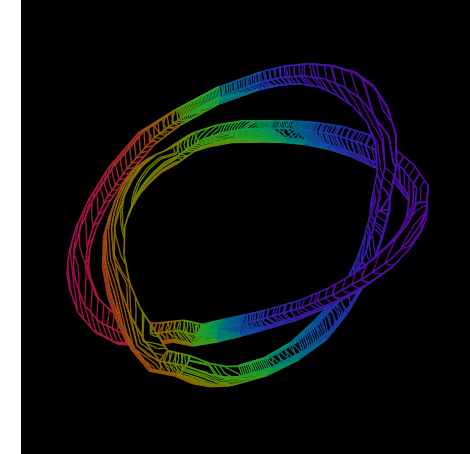
Impulse difference $\cdot k = 12$



Deposited heat / edge flux $\cdot k = 12$



Same final state $\cdot$ entropy level families

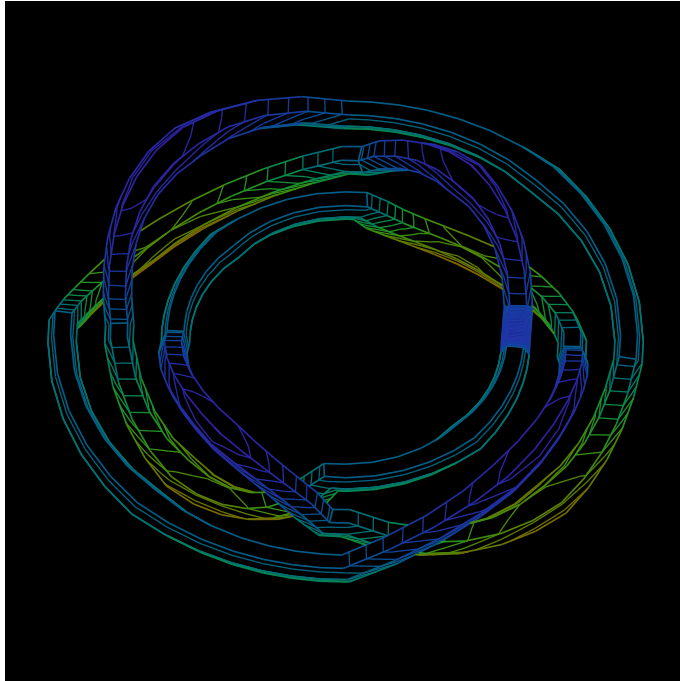


Exact source. Scatter $R = \begin{pmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix}$; attenuate by $d = \frac{15}{16}$; move one incident edge per step. Initial ports are $(1, i)$ at one station, zero elsewhere. Heat receives $(1 - d^2)P_i$ and diffuses through self/neighbor weights $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$. Every step checks total wave energy plus heat = 2. Here $t = \frac{k}{24}$. The lower receivers expose the source-relative impulse, heat/edge flux in units $\frac{2}{N}$, and final cross-entropy. Black omission does not erase the retained carrier.

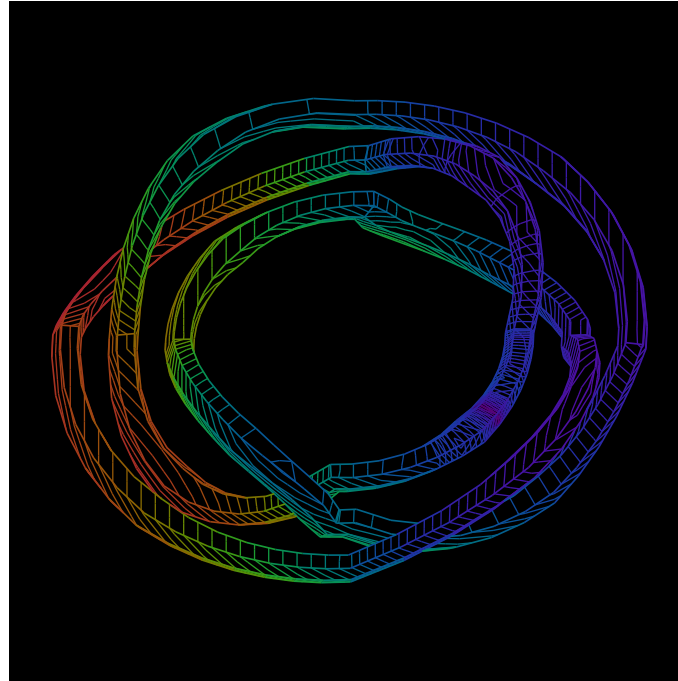
The figure-eight keeps its crossing law through the flow

A different prime carrier, the same constitutive wave law and the same receiver; topology does not supply a new palette.

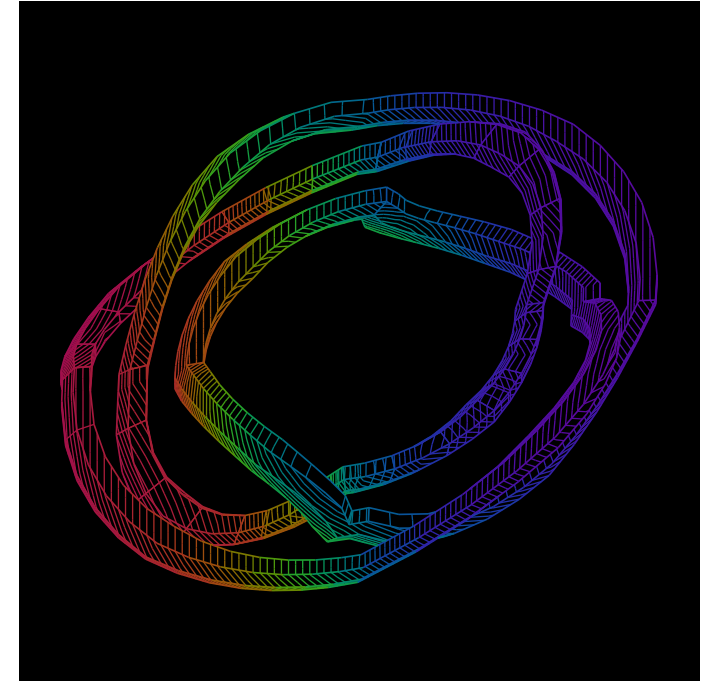
$$k = 0, t = 0$$



$$k = 12, t = \frac{1}{2}$$



$$k = 24, t = 1$$



$$F_t(x, y, z) = \left(x + \frac{tz}{2}, y + \frac{t^2x}{4} + \frac{t^3z}{8}, z \right), \quad \det DF_t = 1$$

$$U_t = \left(\frac{z}{2}, t\frac{x}{2} + t^2\frac{z}{8}, 0 \right), \quad f_t = \left(0, \frac{x}{2} + t\frac{z}{2}, 0 \right), \quad P = 0, \quad \text{curl } U_t = \left(-\frac{t^2}{8}, \frac{1}{2}, \frac{t}{2} \right)$$

$$\partial_t U + (U \cdot \nabla)U = \nu \Delta U - \nabla P + f, \quad \nabla \cdot U = 0, \quad \Delta U = 0$$

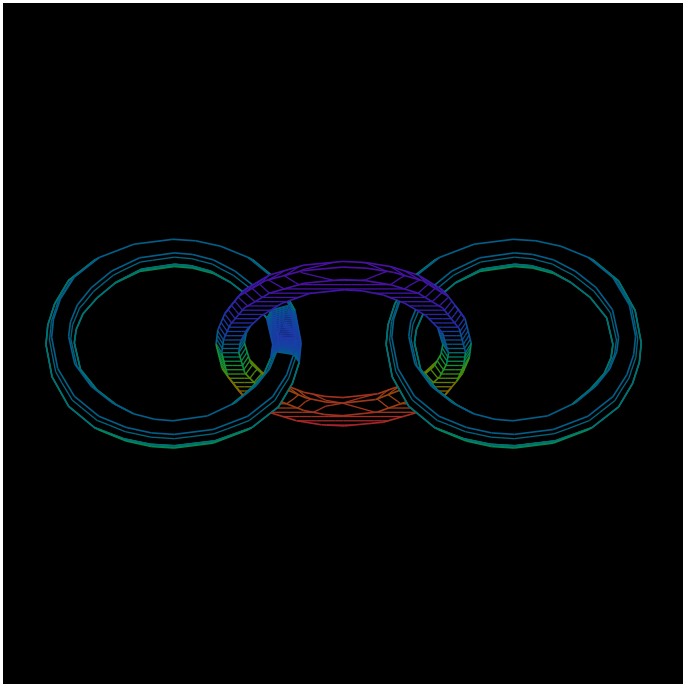
Proved-derived · exact polynomial check. These identities hold holomorphically on $\mathbb{C}^3$. The displayed mesh lies in the invariant real slice. F_t is spatially affine and invertible, so it transports the actual triangular faces and preserves the knot for all real t . The external source f is displayed; this is a forced material-flow realization.

Declared joint receiver. The complex face reads $U + i\ell \text{curl } U + (\Psi^+ + \Psi^-)DF_tv_i$, with material tangent v_i and lane-length chart $\ell = 1$. Velocity, rotation and internal propagation remain separately recoverable. The same coherent primary law gives the colors, with aperture $\frac{1}{16}$.

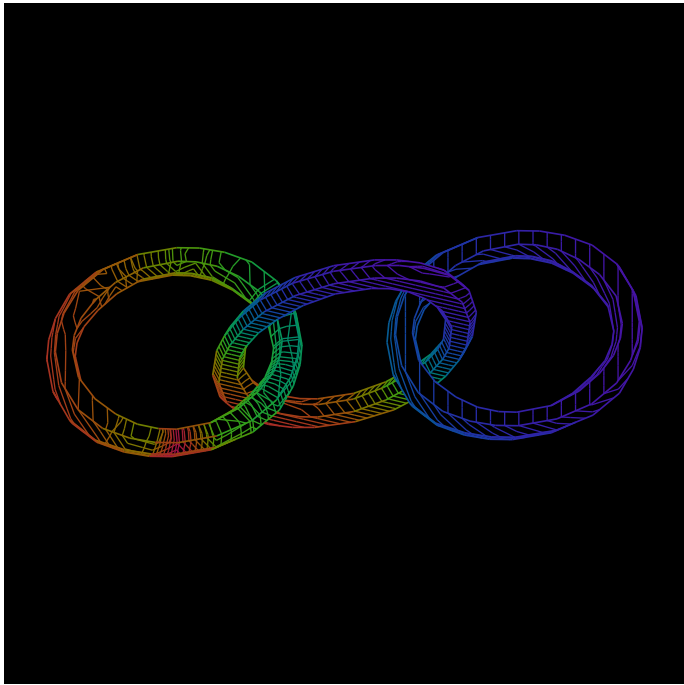
A chain restricts motion before a contact is excited

Three unknotted components retain their linking under the same bulk perturbation; an impulse starts on the first component.

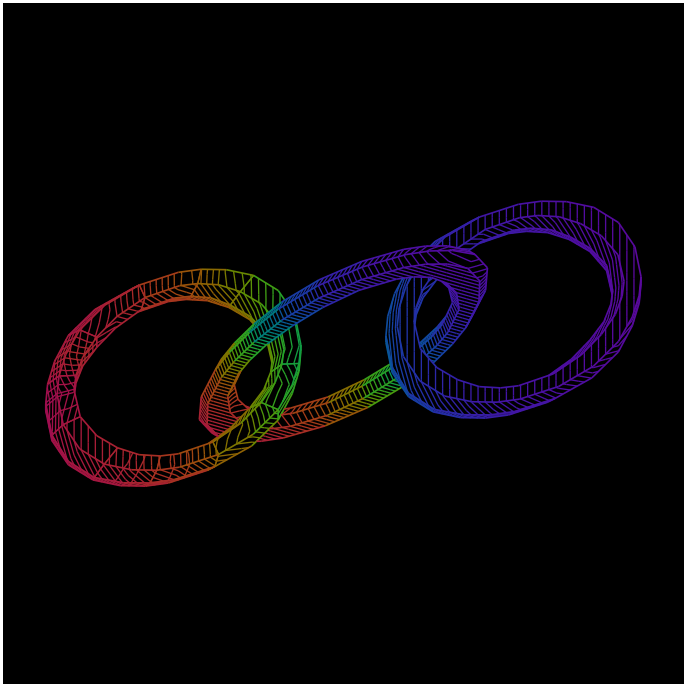
$k = 0, t = 0$



$k = 12, t = \frac{1}{2}$



$k = 24, t = 1$



adjacent pair: Hopf link, end pair: separated, $F_t^{-1}F_t = I$

Linking constrains admissible relative continuation. Established-bounded. Three planar polygonal cores have adjacent spanning-disk intersections. Their closed triangular tubes are disjoint. The common volume-preserving flow carries the chain without crossing. This fixes topology while allowing its receiver shape and orientation to change.

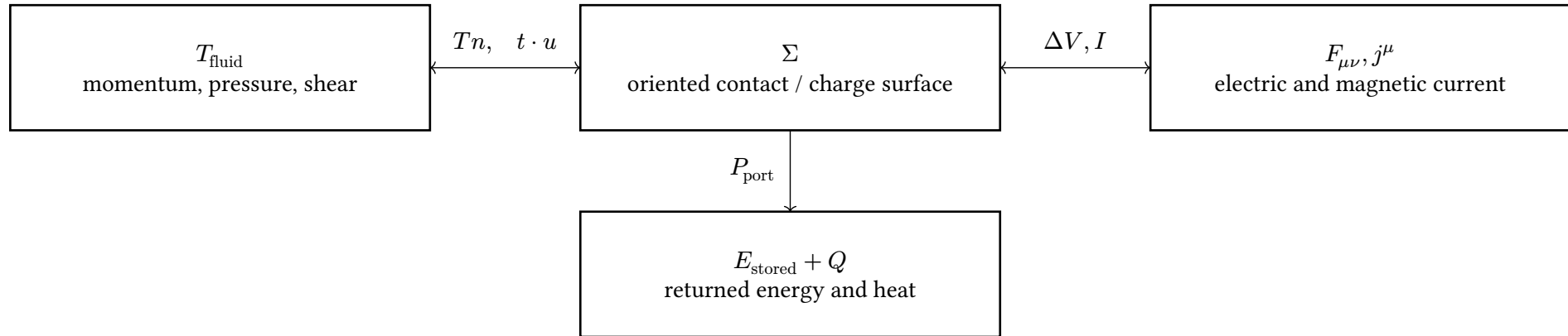
$$Q_i^{k+1} = \frac{Q_{i-1}^k + 2Q_i^k + Q_{i+1}^k}{4} + (1 - d^2)P_i^k, \quad \sum_{i(P_i^k + Q_i^k)} = 2$$

Proved-derived · exact finite passage. A local wave impulse, dissipated heat and receiver cross-entropy have distinct returns. Entropy hatching measures the emitted channel distribution; it does not govern topology or replace the signed wave current. The displayed chain is carried by the prescribed bulk source, not a solved free fluid–elastic-body feedback system.

Contact supplies the actual exchange law. Definition. Internal wave incidence stays on each separated component; an impulse does not jump across a projected crossing. Once an admitted contact is reached, its grip/slip and work ports couple the bodies. Plates 10–11 already derive that exchange, frictional heat and diffusion from a real common patch.

Stress and current cross the same material boundary

Fluid traction, electromagnetic work and capacitive storage meet through their actual ports.



The fluid return retains rotational momentum.

$$\rho D_t u = -\nabla p + \nabla \cdot \tau + j \times B + f, \quad \tau = 2\mu D, \quad D = \frac{\nabla u + (\nabla u)^T}{2}$$

$$\partial_t u = \nu \Delta u + u \times \text{curl } u - \nabla \left(\frac{p}{\rho} + \frac{|u|^2}{2} \right) + \frac{f}{\rho}$$

Conditional physical chart. Incompressible constant-density Newtonian fluid; the second line uses force-inclusive f . Viscous heat is $2\mu D : D$. Maxwell stress adds both traction and energy flow; a surface normal alone supplies neither.

$$E = \frac{1}{2}(Cv^2 + L^{-1}\varphi^2), \quad e = \nabla E, \quad \dot{x} = s - Ge, \quad \dot{E} = e \cdot s - e \cdot Ge, \quad \dot{Q} = e \cdot Ge, \quad \frac{d}{dt}(E + Q) = e \cdot s$$

Proved-derived · Lean. PortEnergyHeat derives the last balance from coordinate derivatives and the storage law. $x = (v, \varphi)$ has conjugate $e = (Cv, L^{-1}\varphi)$; the supplied source and conductance coefficients carry the corresponding units. Contact drag uses the same work/return pattern with its own constitutive law.

The same boundary can store an electrical difference.

$$Q_S = \int_S D_{\text{EM}} \cdot n, \quad I_S = \int_S j \cdot n, \quad \Delta Q = C \Delta V$$

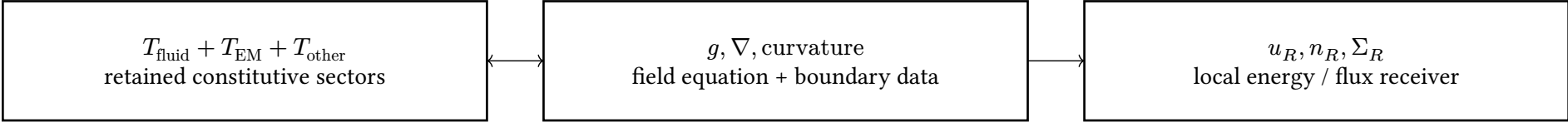
$$j = \sigma(E + u \times B), \quad P_{\text{Ohm}} = \int_V \frac{|j|^2}{\sigma}$$

Conditional circuit reduction. The closed boundary reads enclosed free charge; the charge/voltage comparison also needs the medium and boundary law. Here $\sigma > 0$ is scalar. A changing magnetic flux requires the full path emf. Lightning adds ionization, conductivity evolution and moving leader incidence. The existing leader construction supplies that incidence question.

A receiver reads energy, momentum and tension together

Einstein's source retains energy and stress; the observer changes their measured faces.

$$G^{\mu\nu} + \Lambda g^{\mu\nu} = \frac{8\pi G}{c^4} T_{\text{total}}^{\mu\nu}, \quad \nabla_\mu T_{\text{total}}^{\mu\nu} = 0$$



$$T_{\text{fluid}}^{\mu\nu} = (e + p)u^\mu u^\nu + pg^{\mu\nu} + \pi^{\mu\nu}, \quad g(u, u) = -1 \quad (c = 1, \pi = \text{shear stress})$$

$$J^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}, \quad \partial_\lambda J^{\lambda\mu\nu} = T^{\mu\nu} - T^{\nu\mu} + x^\mu f^\nu - x^\nu f^\mu$$

Proved-derived · Lean. The angular-current identity assumes $\partial_\lambda T^{\lambda\nu} = f^\nu$ and retains antisymmetric stress. Symmetric stress and zero divergence cancel them in the inertial derivative chart. Spin may contribute an additional current; curved spacetime needs the corresponding connection and symmetry/Killing field for a conserved global charge.

A concrete observer change separates dark-sector proposals.

$$T = \text{diag}(e, p_1, p_2, p_3), \quad u_R = \left(\frac{5}{3}, \frac{4}{3}, 0, 0\right), \quad n_R = \left(\frac{4}{3}, \frac{5}{3}, 0, 0\right)$$

$$T(u_R, u_R) = \frac{25e + 16p_1}{9}, \quad T(u_R, n_R) = \frac{20(e + p_1)}{9}$$

Source chart	Energy	Mixed reading
Dust: $p_j = 0$	$\frac{25e}{9}$	$\frac{20e}{9}$
Vacuum: $p_j = -e$	e	0

Tension is a testable stress choice.

$$e + \sum_j p_j = \begin{cases} e & \text{dust} \\ -2e & \text{vacuum} \\ 0 & \text{axial: } p_1 = -e \\ p_2 = p_3 = 0 \end{cases}$$

Interpretation. Investigate dark contributions through density, anisotropic stress, exchange and receiver fibres. Vacuum negative pressure and clustering matter have different transported readings. The physical identifications owe lensing, expansion and structure-formation returns; one scalar “tension” cannot identify all three.

$$E^2 = c^2|p|^2 + (m_0c^2)^2, \quad p = 0, E \geq 0 \rightarrow E = m_0c^2$$

Proved-derived · Lean. The positive rest branch follows from the existing mass-shell owner with $m_0 \geq 0, c > 0$.

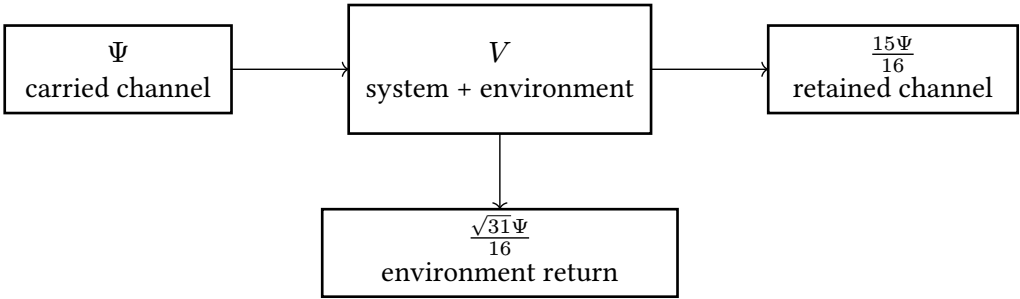
Sources: Tong, General Relativity PDG dark matter PDG dark energy. The existing Einstein–fluid interface derives conservation from the field equation, Bianchi identity and metric compatibility.

Quantum paths retain amplitudes and interaction rules

A knotted material carrier, a circuit graph and a Feynman graph have different edges; their compositional maps can be explicit.

A local complex wave has a reversible completion.

$$R = \begin{pmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix}, \quad R^\dagger R = I$$
$$V = \begin{pmatrix} \frac{15}{16} & -\frac{\sqrt{31}}{16} \\ \frac{\sqrt{31}}{16} & \frac{15}{16} \end{pmatrix}, \quad V^\dagger V = I$$



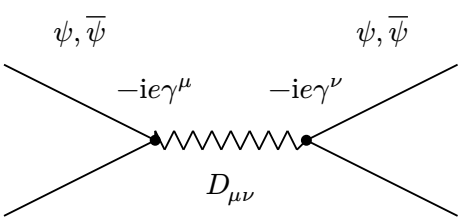
Proved-derived. The same finite scattering map preserves classical complex wave power and quantum amplitude norm. Reading the discarded port as physical heat additionally requires a reservoir and its energy law. $\sqrt{31}$ is derived from $16^2 - 15^2$, not a fitted rendering constant.

Sector	Connection / fibre	What the common carrier preserves
Electromagnetic	$U(1)$	Oriented paths, joins and returned face holonomy
Weak / electroweak	$SU(2)_L \times U(1)_Y$	Sector representation and Higgs coupling remain attached
Strong	$SU(3)$	Color representation, vertices and gauge constraints remain attached
Gravity	Frame / Spin connection	Metric, stress-energy source and causal receiver remain attached

Existing formal carrier. HolonicFourForceSectorCarrier and gauge-covariance owners retain the common incidence and ordered transport; they do not erase the different actions or quantize gravity. A real Lorentzian stress source from complex fields is derived from a real action, with its conjugate channels retained. The holomorphic C^3 fluid chart does not become that physical source merely by taking its real part.

Sources: Tong, Quantum Field Theory, §§1.3, 3.4, 6 Holonics finite quantum transport, Yang–Mills covariance and four-force owners. ETP compares actual substitution/composition laws, beyond a shared graph silhouette.

A propagator and a vertex make the graph executable.



$$G = G_0 + G_0 \Sigma G, \quad G = (G_0^{-1} - \Sigma)^{-1}$$

Conditional. The drawn QED exchange has fermion lines, two coupling vertices and a photon propagator. It is a term in an amplitude expansion, not an electron trajectory or a literal knot crossing. The displayed Dyson relation is an exact inverse identity where the inverses exist; its iterated series also needs convergence or a declared formal-series chart.